Phishing Digital Twin — Demo Walkthrough

How to present the code after opening the Streamlit app

Open with one line

> *"The app is the front end; behind it, a simulated human acts from hidden traits, and a digital twin learns to predict those actions without ever seeing them."*

Part 1 — Show it working (2 min)

1. Open the app at http://localhost:8501
2. Pick a staff member in the sidebar (e.g. Grace Okonkwo, P03)
3. Read one email aloud, click **Yes** or **No**
4. Point to the bar chart updating on the right

   Say: "A real person answers whether this employee would click. The model updates its guess live after every answer, and each answer is saved to JSON."

Part 2 — Trace one click through the code (main part)

Say: "Let me show what happened when I clicked that button." Open the files in this order:

File	What to say
app.py	The button calls three things in order: twin.predict → human.respond → twin.update. That is the whole loop.
hospital_people.json	The 25 people. Each has realistic fields plus a hidden 'simulation' block with hidden_probs — their secret click chance per cue. This is the ground truth.
persona.py	load_people() turns the JSON into objects. Rule: hidden_probs is for the simulated human only.
generator.py	Serves an email. Each email has a cue (urgency, authority, etc.) and a label.
human.py	respond() looks up the person's hidden chance for that cue and rolls a die → click / ignore. situation.py nudges it for night shift or heavy workload.
twin.py (the core)	The predictor. Counts clicks vs not-clicks per cue; guess = clicks / total. predict guesses, update learns. It NEVER reads hidden_probs — only what it observes.

Part 3 — The evidence (1 min)

Run in the terminal: `python3 test_twin_all.py`

Say: "This runs the loop for all 25 people and scores the twin by how close its guesses are to the truth. Accuracy improves with more data — cues seen often are predicted well; rarely-seen cues stay near the 0.5 starting guess."

If the professor asks: 'Why can't the twin see hidden_probs?'

> *"Then it would copy the answer and be perfect — proving nothing. The whole point is predicting from observation only, like a real security team that cannot read minds."*

Spend the most time on **twin.py** and **app.py** — they are the heart. GitHub: github.com/vvek17/Digital_twin